



ITC- Information Technology Colleague



Global Energy Consumption Analysis

Power BI Interactive Dashboard

ONL4_DAT2_S2 • Data Analysis Track • ONL4_DAT2_TEAM 3

Team Members

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AGENDA

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Global Energy Consumption Analysis

Overview

Project Description

This project analyzes global energy consumption and production across multiple countries and energy sources. Using Power BI, we built an interactive dashboard that enables users to explore trends in Petroleum, Coal, Natural Gas, Nuclear, and Renewable Energy from 1980 to present, with forecasting capabilities to 2030.

Tech Stack

- Power BI Desktop
- Power Query (ETL)
- DAX Measures
- GitHub (Version Control)
- CSV Data Sources

The Team



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Data Analyst



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PROBLEM

Problem Statement

Energy data is scattered across multiple sources and formats, making it difficult for policymakers, researchers, and analysts to gain unified insights. Raw datasets contain missing values, inconsistent formats, and lack the visual context needed for informed decision-making. There was no centralized, interactive solution to analyze and compare global energy trends across different sources and countries.

01

Data Fragmentation

Energy data spread across multiple CSV files with no unified structure.

02

Data Quality Issues

Missing values, inconsistent formats, and unclean raw datasets.

03

Lack of Visualization

No interactive way to explore trends, compare countries, or forecast future consumption.

Impact of the Problem

Without a unified analytics solution, stakeholders struggle to identify consumption patterns, compare energy sources, and make data-driven decisions for sustainable energy planning. This leads to delayed insights and missed opportunities for optimizing energy strategies.

Proposed Solution

The Solution

We developed an interactive Power BI dashboard that consolidates global energy data from multiple CSV sources. Using Power Query, we cleaned and transformed the data. DAX measures were created to calculate KPIs and enable dynamic filtering. The dashboard features multiple pages for different energy sources with charts, maps, and forecasting capabilities.

Key Features

- Multi-page interactive dashboard
- Dynamic filtering by country & year
- KPI cards with real-time metrics
- Forecasting to 2030
- Cross-page navigation

Tools & Technologies

Power BI

Data visualization & dashboard creation

Power Query

ETL, data cleaning & transformation

DAX

Custom measures & calculated columns

GitHub

Version control & team collaboration

Expected Outcomes

A fully interactive dashboard enabling users to explore global energy trends, compare countries, analyze consumption patterns, and forecast future energy demands with data-driven insights.

System Design

Data Sources

- Consumption_Total.csv
- Production_Total.csv
- Consumption_Petroleum.csv
- Production_Petroleum.csv
- Consumption_Coal.csv
- Production_Coal.csv
- Consumption_NaturalGas.csv
- Production_NaturalGas.csv
- Oil_price.csv
- consumption_nuclear&renewable.csv

Data Processing

1. Import CSV files into Power Query
2. Remove duplicates & null values
3. Standardize column names
4. Merge related tables
5. Create calculated columns
6. Load to Data Model

Dashboard Pages

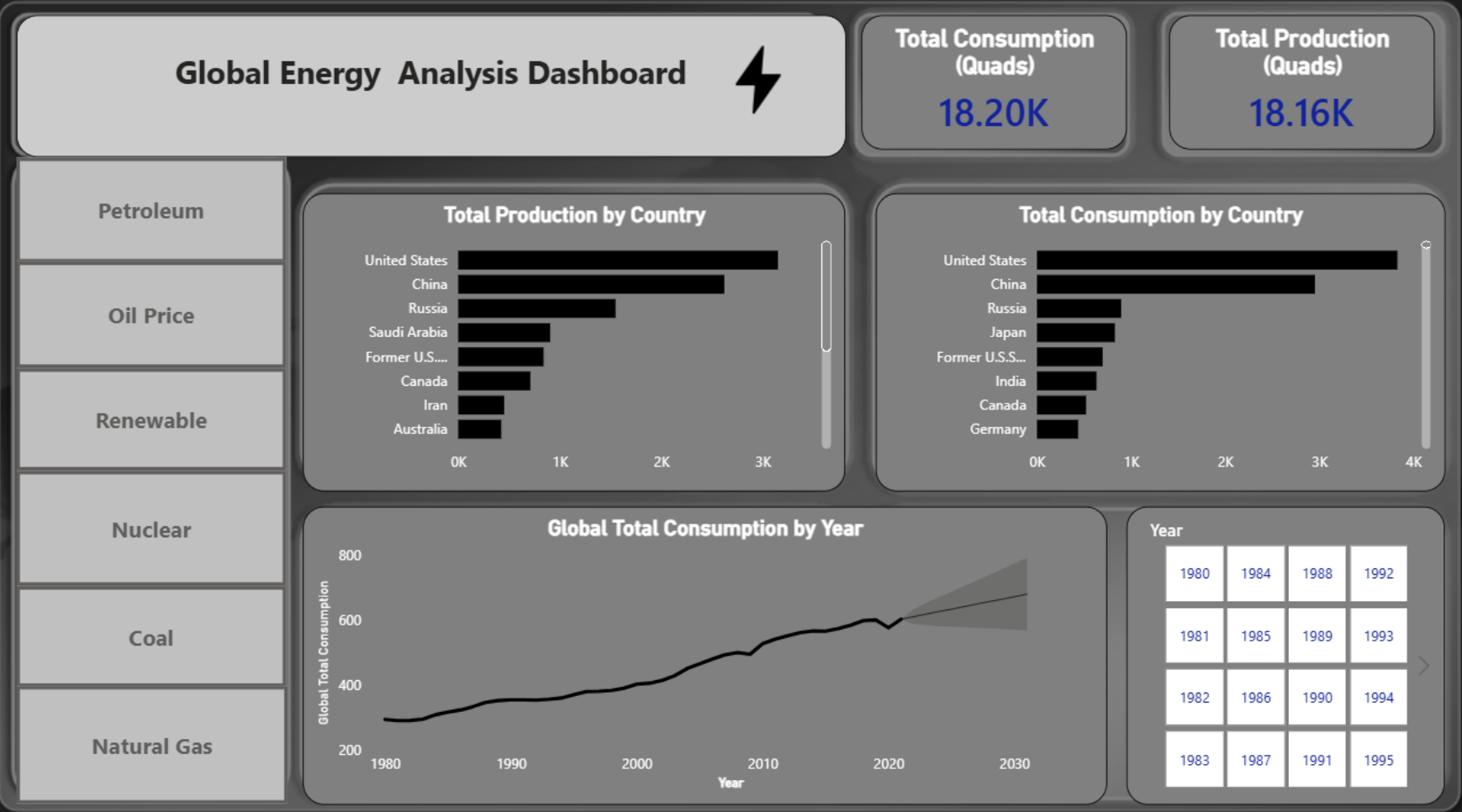
- Global Overview
- Petroleum Analysis
- Oil Price Analysis
- Renewable Energy
- Nuclear Analysis
- Coal Analysis
- Natural Gas Analysis

Data Model Architecture

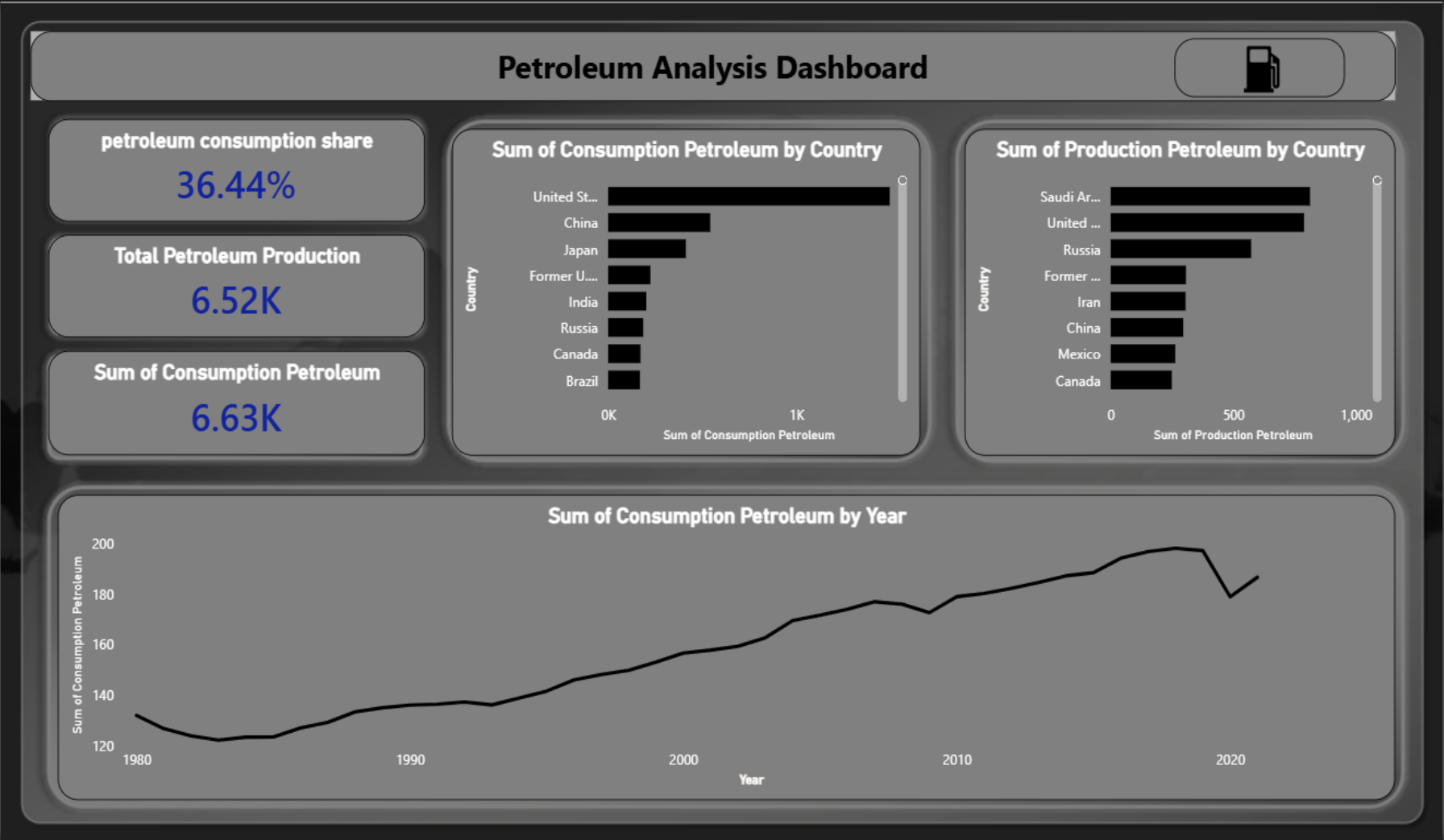
Raw CSV Files → Power Query (ETL) → Cleaned Data Model → DAX Measures → Interactive Dashboard

- Star Schema: Fact tables connected to Dimension tables
- Relationships: One-to-many between dimension and fact tables
- DAX Measures: KPIs, YoY growth, shares, forecasting calculations
- Slicers: Year filter, Country filter, Energy type filter

Global Energy Analysis Dashboard



Petroleum Analysis Dashboard

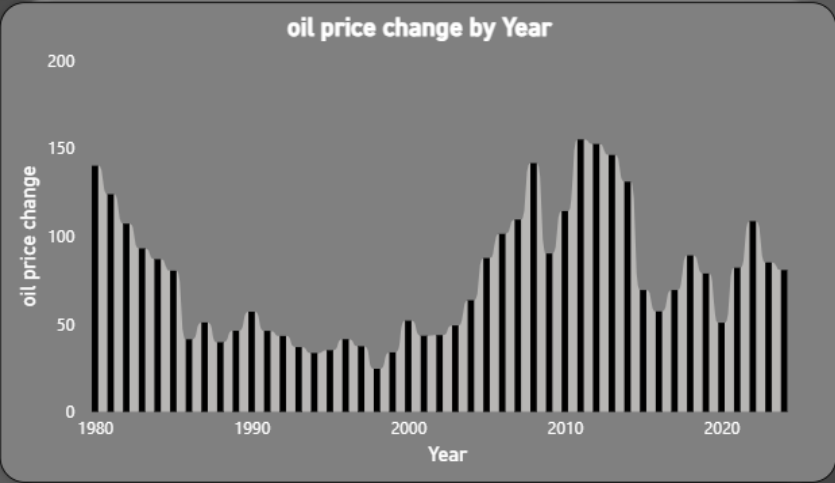
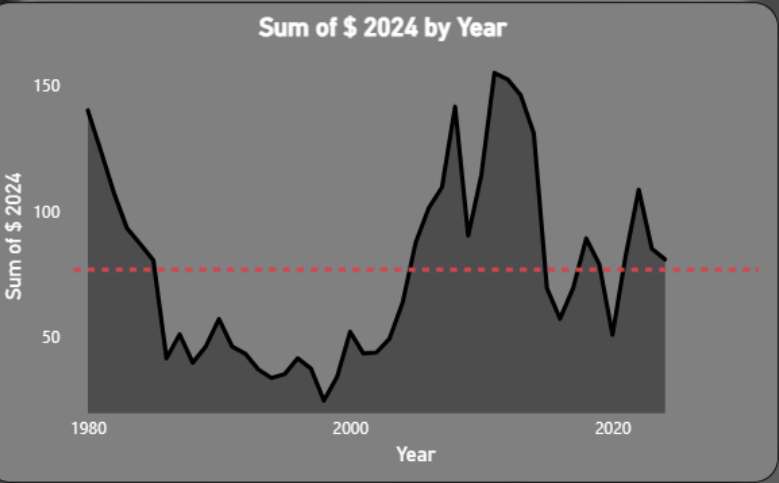
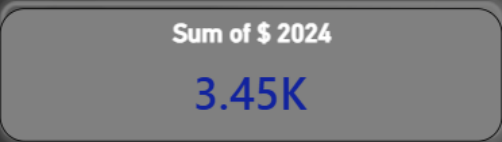
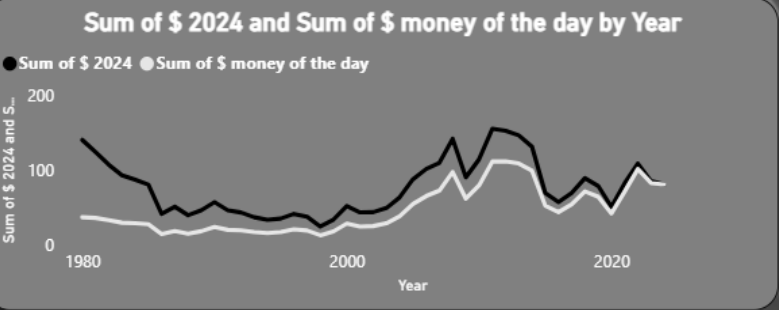


Petroleum Analysis Dashboard

Purpose:
Analyzes petroleum production and consumption worldwide. It compares the top producing and consuming countries, shows petroleum's share of total energy consumption, and tracks consumption trends over time.

Oil Price Analysis Dashboard

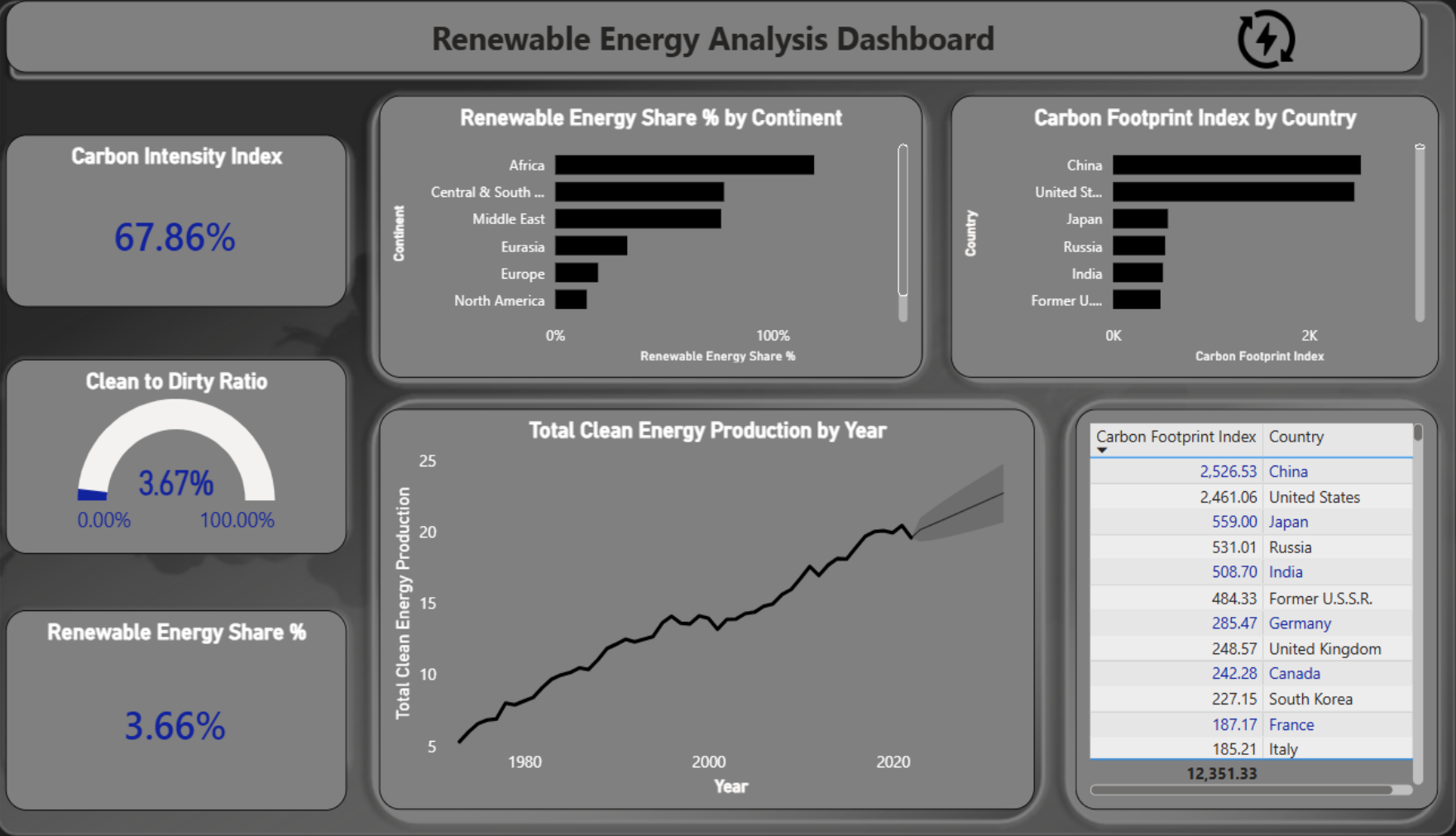
Oil Price Analysis Dashboard



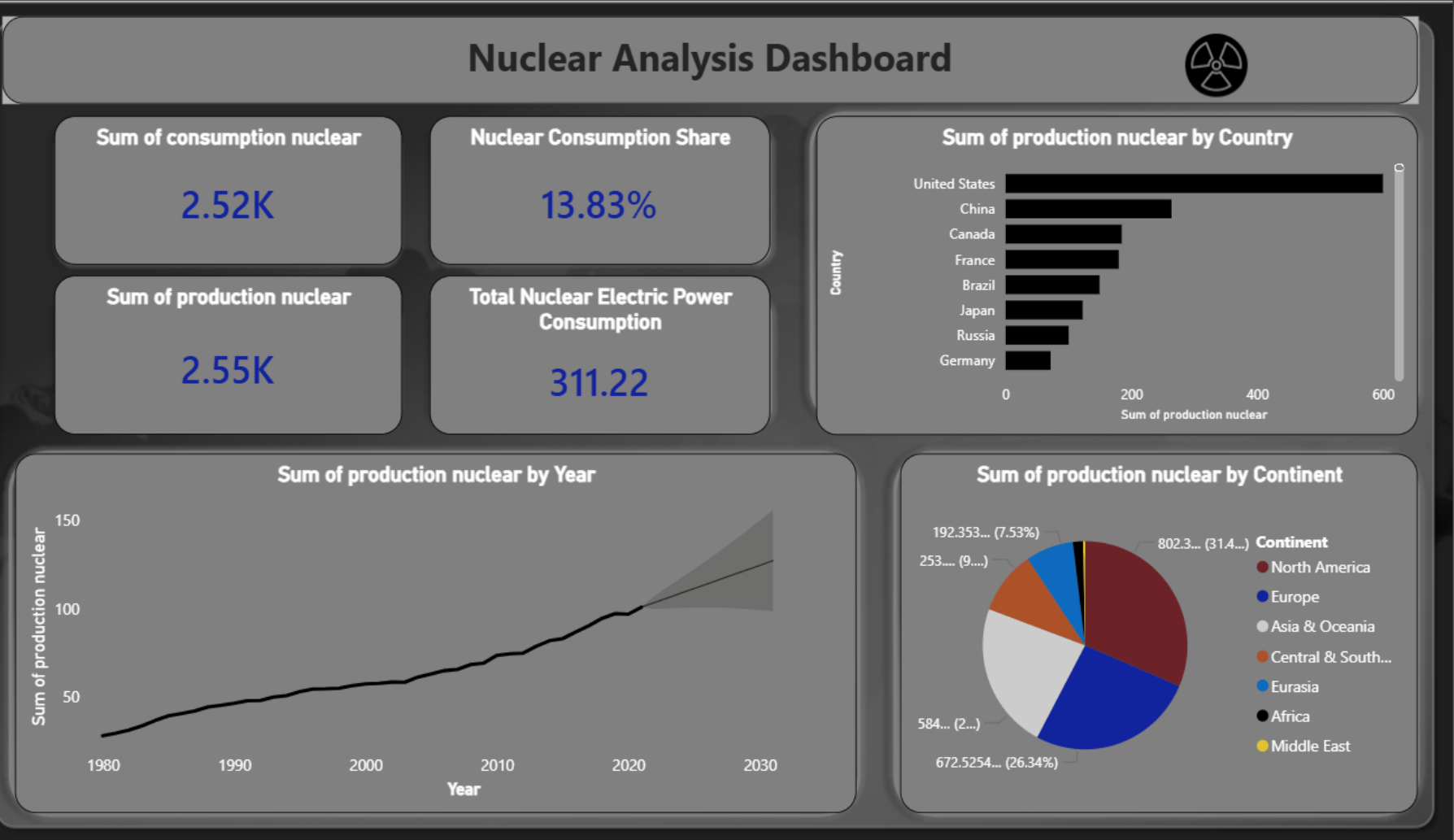
Oil Price Analysis Dashboard Purpose:

Examines historical oil price trends and annual price changes. It displays average oil prices, yearly fluctuations, and helps identify periods of significant price increases or decreases.

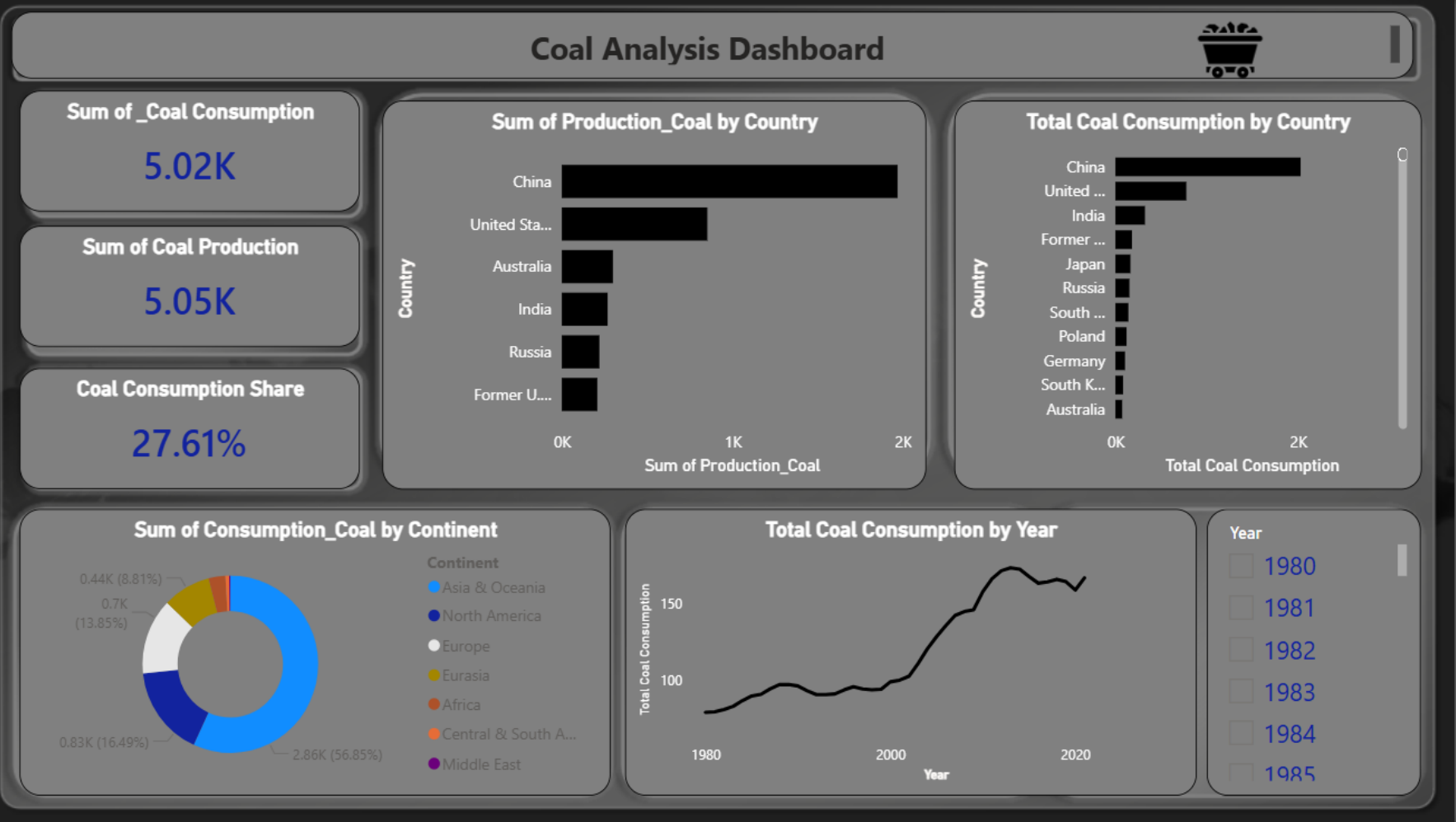
Renewable Energy Analysis Dashboard



Nuclear Analysis Dashboard



Coal Analysis Dashboard



Coal Analysis Dashboard
Purpose:
Analyzes global coal production and consumption. It identifies the leading coal-producing and consuming countries, shows coal's share by continent, and tracks consumption trends over time.

Natural Gas Analysis Dashboard

Natural Gas Analysis Dashboard



Sum of Production_NaturalGas

4.04K

Total Natural Gas Consumption

4.04K

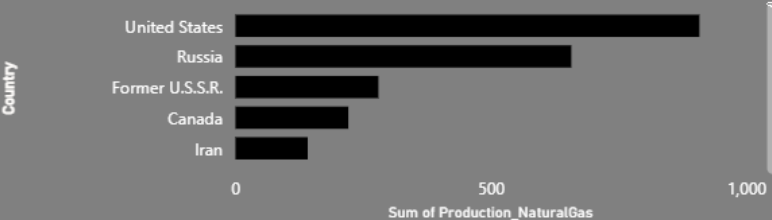
Natural Gas Consumption Share

22.21%

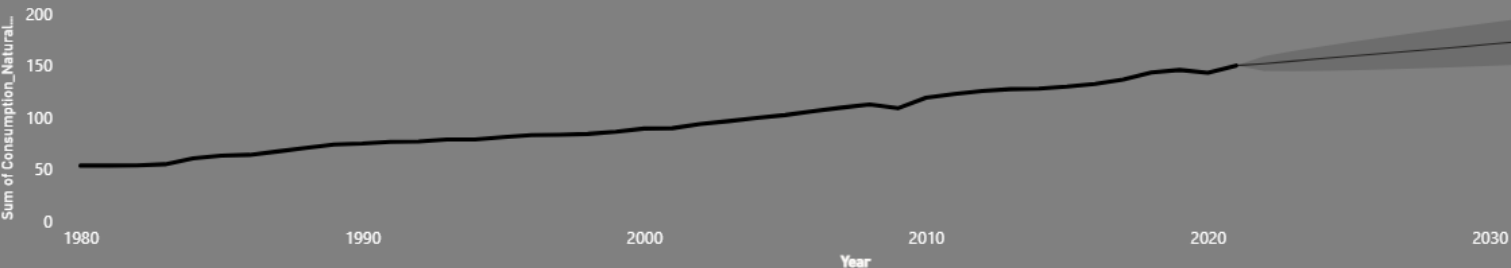
Sum of Consumption_NaturalGas by Country



Sum of Production_NaturalGas by Country



Sum of Consumption_NaturalGas by Year



Natural Gas Analysis Dashboard

Purpose:

Presents insights into natural gas production and consumption. It compares countries by production and consumption, displays natural gas's contribution to global energy use, and visualizes yearly consumption

CHALLENGES

Challenges & Solutions

Data Cleaning

Challenge: Raw datasets contained missing values, inconsistent formats, and duplicates across 12 CSV files.

Solution: Used Power Query to standardize formats, remove duplicates, fill missing values, and create a unified data model.

Data Model Design

Challenge: Combining data from multiple files into a coherent structure with proper relationships.

Solution: Designed a star schema with fact and dimension tables, establishing one-to-many relationships for optimal performance.

DAX Measures

Challenge: Writing complex calculations for KPIs, shares, and forecasting while maintaining performance.

Solution: Created optimized DAX measures with variables and filter context to calculate KPIs efficiently.

Dashboard Performance

Challenge: Large dataset caused slow loading times and lag during interactions.

Solution: Optimized the data model, reduced cardinality, and used aggregation tables to improve response time.

Team Collaboration

Challenge: Coordinating work among 6 team members on the same project files.

Solution: Used GitHub for version control, organized files in repositories, and maintained clear documentation.

Visualization Choice

Challenge: Selecting the most appropriate charts to represent different types of data clearly.

Solution: Researched best practices and chose bar charts for comparisons, line charts for trends, and pie charts for shares.

FUTURE

Future Improvements

01 Real-Time Data Integration

Integrate live energy data from online APIs for up-to-date insights.

02 Forecasting Models

Add advanced forecasting models to predict future energy consumption trends.

03 Power BI Service

Publish the dashboard to Power BI Service for online access and sharing.

04 Expanded Coverage

Include more countries and additional energy-related indicators.

05 Advanced Filtering

Add drill-through pages and advanced filtering for deeper analysis.

06 Performance Optimization

Optimize data model and DAX measures for faster dashboard performance.

07 Row-Level Security

Implement RLS to support secure access for different user roles.

08 Mobile Layout

Create a mobile-friendly dashboard layout for smartphones and tablets.

RESULTS

Key Results

The project successfully transformed raw energy consumption data into an interactive and insightful Power BI dashboard. The dashboard enables users to explore global energy trends through dynamic visualizations and filters.

7

Dashboard Pages

Interactive pages covering all energy sources

12

CSV Files

Raw datasets cleaned and transformed

40+

Years of Data

From 1980 to present with forecasting to 2030

5

Energy Sources

Petroleum, Coal, Natural Gas, Nuclear, Renewable

6

Team Members

Collaborated using GitHub version control

∞

Insights

Data-driven decisions for energy planning

Project Achievements

- Developed an interactive dashboard for analyzing global energy consumption
- Successfully cleaned and transformed the dataset using Power Query
- Created DAX measures to calculate key performance indicators (KPIs)
- Enabled dynamic filtering by country and year for detailed analysis
- Compared different energy sources with clear visualizations
- Improved data accessibility by presenting complex information in a user-friendly format



Thank You!

Global Energy Consumption Analysis

We successfully built an interactive Power BI dashboard that transforms raw energy data into actionable insights. Our solution enables stakeholders to explore trends, compare countries, and forecast future energy demands.

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